

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

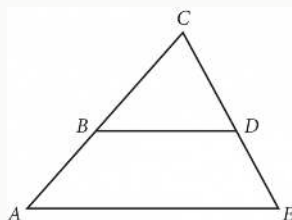
Lines, angles, and triangles

10 questions · ~15 min

02

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62° , what is the measure of angle CAE ?

- A. 58°
- B. 60°
- C. 62°
- D. 120°

Q2

GRID-IN

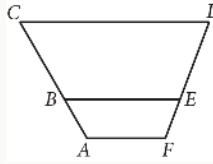
GEOMETRY AND TRIGONOMETRY

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 69° . What is the measure, in degrees, of angle F ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

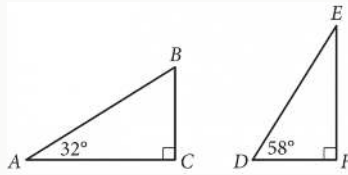


In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

- A. 16.8
- B. 17.5
- C. 18.4
- D. 19.6

In triangle ABC, the measure of angle A is 50° . If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

- A. 50°
- B. 65°
- C. 80°
- D. 100°



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

A.

$$\frac{DE}{DF}$$

B.

$$\frac{DF}{DE}$$

C.

$$\frac{DF}{EF}$$

D.

$$\frac{EF}{DE}$$

Q6

GEOMETRY AND TRIGONOMETRY

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

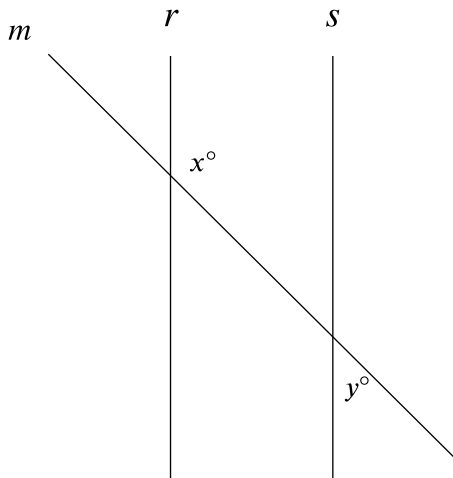
- A. 3
- B. 4
- C. 8
- D. 27

Q7

GEOMETRY AND TRIGONOMETRY

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\frac{k}{2}^\circ$. What is the value of k ?

- A. 36
- B. 45
- C. 72
- D. 108



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled m , r , and s .
- Line m intersects both line r and line s .
- At the intersection of line m and line r , 1 angle is labeled clockwise from top left as follows:
 - Top right: x°
- At the intersection of line m and line s , 1 angle is labeled clockwise from top left as follows:
 - Bottom right: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- A. $x < 115$
- B. $x > 115$
- C. $x + y < 180$

D. $x + y > 180$

Q9

GEOMETRY AND TRIGONOMETRY

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

A. 10°

B. 30°

C. 40°

D. 90°

Triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC . The measure of side AB is 16. What is the measure of side XY ?

- A. 14
- B. 16
- C. 18
- D. 32